

LONGLONG XU

Multimodal LLMs; Post-Training and Alignment; Multimodal Reasoning; LLMs for AIOps

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🎓 EDUCATION

Tsinghua University, Department of Computer Science and Technology Sep. 2023 - Present

Ph.D. Student Computer Science and Technology

Research Interests: Multimodal LLMs, post-training and alignment, multimodal reasoning, LLMs for AIOps

Tsinghua University, Department of Computer Science and Technology Sep. 2018 - Jun. 2023

B.Eng. Computer Science and Technology

🧑‍🔬 REPRESENTATIVE RESEARCH

- **LLM Post-Training and Reasoning**: two ongoing first-author works on intent-aligned time-series anomaly detection and multimodal LLM reasoning over long numerical sequences; **FoundRoot**, ICSE 2026 (CCF A, co-author, [link](#)); **ChatTS**, VLDB 2025 (CCF A, 4th author, [link](#)).
- **LLM Evaluation**: **OpsEval**, FSE 2025 (CCF A, 3rd author, [link](#)); **Eagle**, FSE 2026 (CCF A, 4th author, [link](#)); **TechSupportEval**, IJCNN 2025 (THU B, CCF C, 4th author, [link](#)).
- **Fault Diagnosis**: one ongoing first-author work on software-change-aware fault diagnosis for online service systems; **LatentScope**, KDD 2024 (CCF A, co-author, [link](#)).
- **Time-Series Forecasting**: **TameR**, ICML 2026 (CCF A, first author); **SPRINT**, ICML 2026 (CCF A, first author).
- **Computer Vision**: CVPR 2023 Highlight (CCF A, 3rd author, [link](#)); IJCV 2024 (CCF A, 2nd author, [link](#)).

LLM Post-Training and Applied Reasoning Sep. 2023 - May 2026

- Proposed two ongoing first-author LLM post-training works: one targets human-intent-aligned time-series anomaly detection using data profiling, intent parsing, memory retrieval, and reinforcement learning; the other targets numerical accuracy and fine-grained localization over long sequences via active perception and post-training on multi-turn tool-use trajectories.
- Contributed to **FoundRoot** and **ChatTS**: FoundRoot builds a foundation reasoning model for fault root-cause localization with structured deep thinking; ChatTS introduces time series as a new modality for LLM-based numerical question answering and reasoning.

LLM Evaluation and Benchmark Generation Nov. 2023 - Apr. 2026

- Contributed to **OpsEval**, **Eagle**, and **TechSupportEval**, building benchmarks and automated evaluators for complex reasoning, vertical-domain question answering, operations documents, and technical support.
- Covered alerts, logs, metrics, traces, configurations, and technical documents, evaluating evidence understanding, root-cause analysis, document use, question generation, key-term matching, and step completeness.

Fault Diagnosis and Root-Cause Localization Models Apr. 2024 - Apr. 2025

- Proposed an ongoing first-author method for software-change-aware diagnosis in online services, using change-aware distribution alignment and contrastive fault clustering to maintain stable latent fault representations under concept drift.
- Researched **OTN-STFM** at Huawei, unifying optical-network performance metrics, alarms, and topology with a self-supervised spatio-temporal autoencoder; achieved 91.32%/92.98%/94.21% A@1/A@3/A@5 on 242 fault cases and was practically validated on internal backbone-network fault-localization tasks.

Robust Training and Inference-Time Scaling for Time-Series Forecasting May 2025 - May 2026

- Proposed **TameR**, identifying shortcut reliance on recent observations in forecasting models and improving robustness to noise, missing values, and abnormal perturbations with base-space aligned random sampling, learnable period extraction, and two-stage training.
- Proposed **SPRINT**, an inference-time scaling framework for long and ultra-long forecasting; without retraining or modifying the base model, it improved accuracy by 19%, reduced peak memory by 6.4×, and reduced inference time by 16.9×.

INTERNSHIPS

ByteDance - Algorithm Engineer

May 2025 - May 2026

- Conducted research on LLM post-training, multimodal reasoning, and inference-time enhancement, producing two first-author ICML 2026 papers (**TameR** and **SPRINT**) and two first-author ongoing works.
- Built training and evaluation pipelines for multi-turn reasoning tasks, covering synthetic data generation, supervised fine-tuning, reinforcement learning / trajectory-level alignment, inference service evaluation, and multi-model baselines.
- A related intent-aligned anomaly-detection line has been validated in an internal intelligent alerting workflow.

Huawei - Algorithm Engineer

Oct. 2024 - Apr. 2025

- Researched **OTN-STFM** for optical transport networks, jointly modeling performance metrics, alarms, and topology with a self-supervised spatio-temporal autoencoder for anomaly detection, fault localization, and degradation detection.
- Achieved 91.32%/92.98%/94.21% A@1/A@3/A@5 on 242 fault cases, was practically validated on internal backbone-network fault-localization tasks, and explored channel-aware clustered graph autoencoders for zero-shot localization across topologies, variable counts, and domains.

ZTE - Algorithm Engineer

Nov. 2023 - Sep. 2024

- Collected and cleaned network-domain corpora, generated question-answering data, and fine-tuned general LLMs into an operations-domain LLM, improving domain QA and retrieval-augmented QA performance.
- Contributed to **OpsEval** (FSE 2025, CCF A, 3rd author) and **TechSupportEval** (IJCNN 2025, THU B, CCF C, 4th author), building evaluation data and automated QA evaluators for internal practical LLM development and assessment. (**OpsEval**; **TechSupportEval**)

eBay - Algorithm Engineer

May 2023 - Oct. 2023

- Developed root-cause localization algorithms for complex internal systems by modeling causal relations among system metrics in latent space and using intervention-based identification; contributed to a KDD 2024 (CCF A) co-author paper and internal engineering validation. ([link](#))

RealAI - Algorithm Engineer

Aug. 2022 - May 2023

- Designed adversarial textured 3D meshes to bypass face-recognition defenses; published at CVPR 2023 (CCF A, 3rd author) and selected as a CVPR Highlight (top 2.5%). ([link](#))
- Implemented robust physical face attacks with face patches by perturbing the 3D StyleGAN latent space; published in IJCV 2024 (CCF A, 2nd author). ([link](#))

AWARDS

Outstanding Undergraduate Thesis, Tsinghua University

Jun. 2023

Xiaomi Scholarship

Dec. 2022

Comprehensive Excellence Scholarship, Tsinghua University

Dec. 2021, Dec. 2022

Social Work Excellence Scholarship, Tsinghua University

Dec. 2020

SKILLS

- Programming: Python, C/C++, Java
- Post-Training and LLM Tools: PyTorch, HuggingFace, vLLM, Open-R1, TRL, TensorFlow
- Systems and Engineering: Kubernetes, Docker, Node.js, Django, MongoDB, MySQL, Android, $\LaTeX$